Teaching Case: A Hybrid Fuzzy AHP-TOPSIS Approach for Multi-Criteria Candidate Selection Process

Tran Quoc Dat Nguyen
10422017@student.vgu.edu.vn
Computer Science and Engineering
Vietnamese-German University
Ho Chi Minh, Vietnam

Minh Giap Nguyen
10422024@student.vgu.edu.vn
Computer Science and Engineering
Vietnamese-German University
Ho Chi Minh, Vietnam

Khoi Nguyen Nguyen
10422058@student.vgu.edu.vn
Computer Science and Engineering
Vietnamese-German University
Ho Chi Minh, Vietnam

***Abstract*—Modern students and young professionals are increasingly confronted with complex decision-making scenarios, such as identifying suitable career paths, selecting scholarship opportunities, or evaluating academic and professional development options. These decisions often involve multiple criteria that are subjective, uncertain, and difficult to quantify using traditional evaluation methods. To address this challenge, this project proposes a structured decision-support framework that integrates Fuzzy Analytic Hierarchy Process (Fuzzy AHP) and Fuzzy Technique for Order Preference by Similarity to Ideal Solution (Fuzzy TOPSIS). The system is designed to assist in multi-criteria decision-making across domains such as job position suitability, scholarship selection, and other opportunity assessments. By incorporating fuzzy logic, the model effectively captures ambiguity in human judgment when evaluating criteria such as skills, experience, academic performance, financial need, or personal preferences. Fuzzy AHP is utilized to determine the relative importance of decision criteria, while Fuzzy TOPSIS ranks alternatives by measuring their closeness to an ideal solution. Through this integrated approach, the project provides a quantitative yet flexible mechanism for grading and prioritizing options, enabling individuals and institutions to make more informed, transparent, and rational decisions. The framework ultimately aims to enhance fairness and effectiveness in selection and recommendation processes where multiple qualitative and quantitative factors must be considered simultaneously.**

***Index Terms*—Multi-Criteria Decision Making, Analytic Hierarchy Process, Fuzzy AHP, TOPSIS, Fuzzy TOPSIS, Hybrid Fuzzy AHP-TOPSIS, Candidate Selection, Triangular Fuzzy Numbers, Decision Support System**

Relevant code for implementing this project can be found at: *https://github.com/KhoiNguyenVGU/ahp_calculator*

The web application for this project is available at: *https://ahp-calculator-puce.vercel.app/*

Web-App Tutorial is available on YouTube at: *https://www.youtube.com/watch?v=QfCC6G9wTXk&t=57s*

I. Case Introduction

Complex decision-making is a common challenge in many real-world contexts such as career planning, scholarship selection, and candidate evaluation [1]. Decision-makers are often required to compare multiple alternatives based on a set of diverse criteria that may include both quantitative measurements and qualitative judgments. The evaluation process becomes particularly difficult when subjective perceptions and uncertainty influence how alternatives are assessed. As discussed by Radbruch and Schiprowski [2], evaluation outcomes can be significantly shaped by the way individuals interpret and judge information during the decision process, highlighting the role of subjective assessments in decision-making.

Because of these subjective elements, traditional decision-making methods that rely purely on precise numerical data may not adequately capture the ambiguity present in human judgment. This limitation has encouraged the development of analytical approaches capable of incorporating uncertainty and linguistic evaluations. Among these approaches, fuzzy-based multi-criteria decision-making (MCDM) methods have gained considerable attention for their ability to model imprecision and vagueness in real-world decision problems.

In this project, fuzzy multi-criteria decision-making techniques are applied to support structured evaluation and ranking of alternatives. The framework allows decision-makers to identify relevant criteria, determine their relative importance, and assess candidate performance using fuzzy logic to represent uncertain or subjective judgments. By combining structured evaluation with fuzzy reasoning, the approach improves transparency and consistency in complex decision environments.

The main objective of this project is to demonstrate how fuzzy decision-making models can assist in solving practical evaluation problems. Through the implementation of methods such as Fuzzy Analytic Hierarchy Process (Fuzzy AHP) and Fuzzy Technique for Order Preference by Similarity to Ideal Solution (Fuzzy TOPSIS), learners can gain practical experience in defining decision criteria, assigning weights, evaluating alternatives, and interpreting the resulting rankings.

By decomposing complex evaluation tasks into systematic analytical steps, this framework supports more informed and rational decision-making. Ultimately, the project illustrates how fuzzy multi-criteria analysis can be applied to assist individuals and organizations in making better decisions when dealing with uncertainty and subjective judgments.

II. Literature Review

Multi-criteria decision-making (MCDM) has been widely applied in domains where complex evaluation problems involve both qualitative and quantitative factors. Traditional decision-making methods often struggle to manage subjective judgments, uncertainty, and conflicting criteria, particularly in areas such as personnel selection, scholarship allocation, and career planning.

Among MCDM approaches, the Analytic Hierarchy Process (AHP), introduced by Saaty [3], is one of the most widely used techniques for determining the relative importance of decision criteria through pairwise comparisons. AHP provides a structured framework that decomposes complex problems into hierarchical levels, enabling systematic evaluation of alternatives [4]. However, classical AHP assumes precise numerical judgments, which may not accurately reflect real-world decision-making where human opinions are often vague and uncertain.

To address this limitation, fuzzy extensions of AHP have been developed. Fuzzy AHP integrates fuzzy set theory to capture ambiguity in human judgment, allowing decision-makers to express preferences using linguistic variables rather than exact values [5]. This approach enhances the reliability of weight determination in situations where criteria importance cannot be precisely defined.

For ranking alternatives, the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is frequently employed. TOPSIS evaluates alternatives based on their distance from an ideal and a negative-ideal solution [6]. Similar to AHP, conventional TOPSIS requires crisp numerical inputs, which limits its effectiveness in uncertain decision environments. Fuzzy TOPSIS extends the method by incorporating fuzzy logic, enabling more realistic modeling of subjective evaluations [7].

Recent studies suggest that integrating Fuzzy AHP and TOPSIS provides a robust hybrid framework for solving complex selection problems. Fuzzy AHP is typically used to determine criteria weights, while TOPSIS ranks alternatives based on their closeness to the ideal solution. This hybrid approach has been successfully applied in areas such as personnel recruitment, supplier selection, and project prioritization [8], [1].

Despite the growing adoption of fuzzy MCDM techniques, there remains a need for structured frameworks that support decision-making in academic and professional opportunity selection contexts [9]. This case builds upon existing research by applying a hybrid Fuzzy AHP–TOPSIS model to support systematic evaluation and ranking of candidates or opportunities under uncertain and multi-criteria conditions.

III. Methodology

This section presents the five multi-criteria decision-making (MCDM) methods employed in this teaching case: AHP, Fuzzy AHP, TOPSIS, Fuzzy TOPSIS, and the hybrid approach combining Fuzzy AHP and TOPSIS. Each method is described in terms of its mathematical formulation and computational procedure.

A. Analytic Hierarchy Process (AHP)

1) Classical AHP: The Analytic Hierarchy Process (AHP), introduced by Saaty [3], is a structured technique for organizing and analyzing complex decisions based on pairwise comparisons. The method uses Saaty's fundamental scale of 1–9 to express the relative importance between pairs of criteria or alternatives, where 1 indicates equal importance and 9 indicates extreme importance. Reciprocal values (e.g., $1/3$, $1/5$) represent inverse relationships.

Step 1: Construct the pairwise comparison matrix. Given n criteria, the decision maker constructs an $n \times n$ matrix $A = [a_{ij}]$, where a_{ij} represents the relative importance of criterion i over criterion j. The matrix satisfies $a_{ii} = 1$ and $a_{ji} = 1/a_{ij}$.

Step 2: Compute the priority vector using the geometric mean method. The geometric mean of each row is calculated as:

$$\bar{g}_i = \left(\prod_{j=1}^{n} a_{ij} \right)^{1/n} \tag{1}$$

The priority weights are then obtained by normalizing:

$$w_i = \frac{\bar{g}_i}{\sum_{k=1}^{n} \bar{g}_k} \tag{2}$$

Step 3: Check consistency. To ensure the judgments are logically consistent, the consistency ratio (CR) is computed. First, the principal eigenvalue $\lambda_{\max}$ is estimated as:

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^{n} \frac{(A \cdot \mathbf{w})_i}{w_i} \tag{3}$$

The consistency index is $CI = (\lambda_{\max} - n)/(n - 1)$, and the consistency ratio is $CR = CI/RI$, where RI is the random index for a matrix of size n. A threshold of $CR < 0.10$ is generally accepted [3].

Step 4: Aggregate final scores. When alternatives are also compared pairwise under each criterion, the final score of alternative a is computed as the weighted sum:

$$S_a = \sum_{c=1}^{n} w_c \cdot s_{ca} \tag{4}$$

where s_{ca} is the local priority of alternative a under criterion c.

2) Fuzzy AHP: Fuzzy AHP extends the classical AHP by representing pairwise comparison judgments as triangular fuzzy numbers (TFNs) rather than crisp values, thereby capturing the inherent vagueness in human assessments [5], [10]. A TFN is denoted as $\tilde{a} = (l, m, u)$, where l, m, and u represent the lower, middle, and upper bounds, respectively.

Crisp Saaty scale values are mapped to TFNs using a predefined fuzzy scale. For example, a judgment of 1 maps to $(1, 1, 1)$, a judgment of 3 maps to $(2, 3, 4)$, and a judgment of 9 maps to $(9, 9, 9)$. For reciprocal values, the inverse TFN is used: $\tilde{a}^{-1} = (1/u, 1/m, 1/l)$ [11].

The key fuzzy arithmetic operations on TFNs are defined as:

$$\tilde{a} \oplus \tilde{b} = (l_a + l_b,\ m_a + m_b,\ u_a + u_b) \tag{5}$$

$$\tilde{a} \otimes \tilde{b} = (l_a \cdot l_b,\ m_a \cdot m_b,\ u_a \cdot u_b) \tag{6}$$

Step 1: Build the fuzzy comparison matrix. Convert the crisp pairwise comparison matrix into a fuzzy matrix $\tilde{A} = [\tilde{a}_{ij}]$ using the fuzzy scale mapping.

Step 2: Compute fuzzy geometric means. For each row i:

$$\tilde{r}_i = \left(\prod_{j=1}^{n} \tilde{a}_{ij} \right)^{1/n} \tag{7}$$

where the n-th root is applied component-wise to the TFN.

Step 3: Compute fuzzy weights. The fuzzy weight of criterion i is:

$$\tilde{w}_i = \tilde{r}_i \otimes (\tilde{r}_1 \oplus \tilde{r}_2 \oplus \cdots \oplus \tilde{r}_n)^{-1} \tag{8}$$

Step 4: Defuzzify. The crisp weight is obtained using the Center of Area (CoA) method:

$$w_i = \frac{l_i + m_i + u_i}{3} \tag{9}$$

The crisp weights are then normalized to sum to unity. Final alternative scores are computed identically to classical AHP using these normalized weights.

B. TOPSIS

1) Classical TOPSIS: The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), proposed by Hwang and Yoon [6], ranks alternatives based on their geometric distance from the ideal best and ideal worst solutions. The procedure operates on a decision matrix $X = [x_{ij}]$ where x_{ij} is the performance of alternative i on criterion j.

Step 1: Normalize the decision matrix using vector normalization:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \tag{10}$$

Step 2: Construct the weighted normalized matrix:

$$v_{ij} = w_j \cdot r_{ij} \tag{11}$$

where w_j is the weight of criterion j.

Step 3: Determine the ideal best (V^+) and ideal worst (V^-) solutions. For benefit criteria, $V_j^+ = \max_i(v_{ij})$ and $V_j^- = \min_i(v_{ij})$. For cost criteria, the definitions are reversed.

Step 4: Calculate separation measures using Euclidean distance:

$$S_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - V_j^+)^2} \tag{12}$$

$$S_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - V_j^-)^2} \tag{13}$$

Step 5: Compute performance scores (closeness coefficient):

$$P_i = \frac{S_i^-}{S_i^+ + S_i^-} \tag{14}$$

Alternatives are ranked in descending order of P_i, where a higher score indicates greater proximity to the ideal solution.

2) Fuzzy TOPSIS: Fuzzy TOPSIS extends the classical method by incorporating fuzzy numbers to handle imprecise or uncertain data in the evaluation process [7]. In this implementation, the decision matrix values are represented as TFNs $\tilde{x}_{ij} = (l_{ij}, m_{ij}, u_{ij})$, and the criteria weights are also expressed as TFNs derived from linguistic variables.

Step 1: Construct the fuzzy decision matrix. Crisp performance values x_{ij} are converted to TFNs of the form (x_{ij}, x_{ij}, x_{ij}).

Step 2: Normalize the fuzzy matrix using vector normalization based on the middle values:

$$\tilde{r}_{ij} = \frac{\tilde{x}_{ij}}{\sqrt{\sum_{i=1}^{m} m_{ij}^2}} \tag{15}$$

where the division is applied component-wise to (l, m, u).

Step 3: Compute the weighted normalized fuzzy matrix:

$$\tilde{v}_{ij} = \tilde{w}_j \otimes \tilde{r}_{ij} \tag{16}$$

Step 4: Identify the fuzzy ideal best ($\tilde{F}^+$) and fuzzy ideal worst ($\tilde{F}^-$) based on the defuzzified middle values of the weighted normalized matrix, applying the same benefit/cost logic as in classical TOPSIS.

Step 5: Calculate separation measures using the vertex distance between TFNs:

$$d(\tilde{a}, \tilde{b}) = \sqrt{\frac{1}{3}\left[(l_a - l_b)^2 + (m_a - m_b)^2 + (u_a - u_b)^2\right]} \tag{17}$$

The total distance of alternative i from the ideal best and worst solutions is:

$$d_i^+ = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{F}_j^+), \quad d_i^- = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{F}_j^-) \tag{18}$$

Step 6: Compute the closeness coefficient identically to classical TOPSIS:

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{19}$$

Alternatives are ranked in descending order of CC_i.

C. Hybrid Fuzzy AHP-TOPSIS

The Hybrid Fuzzy AHP-TOPSIS method integrates the strengths of both Fuzzy AHP and TOPSIS into a two-phase framework [12], [13]. Fuzzy AHP excels at deriving reliable criteria weights through structured pairwise comparisons, while TOPSIS is well-suited for ranking alternatives against multiple criteria using distance-based aggregation. By combining both methods, the hybrid approach addresses the limitations of using either method in isolation: Fuzzy AHP alone becomes computationally prohibitive when the number of alternatives is large (requiring pairwise comparisons among all alternatives under each criterion), and TOPSIS alone lacks a systematic mechanism for determining criteria weights from expert judgments.

Phase 1: Fuzzy AHP – Determine criteria weights. The decision maker constructs a pairwise comparison matrix for the n criteria using Saaty's scale. This matrix is converted into a fuzzy comparison matrix $\tilde{A} = [\tilde{a}_{ij}]$ using the TFN mapping described in Section III-A2. The fuzzy geometric mean method is then applied to compute fuzzy weights $\tilde{w}_j$ for each criterion, following Steps 1–4 of Fuzzy AHP. The resulting fuzzy weight vector $\tilde{\mathbf{w}} = (\tilde{w}_1, \tilde{w}_2, \ldots, \tilde{w}_n)$ is carried forward to Phase 2.

Phase 2: TOPSIS – Rank alternatives. A decision matrix $X = [x_{ij}]$ of m alternatives evaluated on n criteria is constructed directly from performance data (rather than pairwise comparisons). The TOPSIS procedure described in Section III-B1 is then executed, using the fuzzy weight vector $\tilde{\mathbf{w}}$ from Phase 1 in Step 3 to compute the weighted normalized matrix. The closeness coefficients CC_i yield the final ranking of alternatives.

This two-phase design makes the hybrid method particularly suitable for scenarios with a moderate number of criteria but a large number of alternatives, as is common in candidate selection problems.

IV. Data Source

A. Kaggle Dataset

The dataset used in this teaching case is the *Interview Selection Dataset*, publicly available on Kaggle (*https://www.kaggle.com/datasets/suryasanju/interview-selection-dataset*). It was originally collected from a large-scale recruitment process conducted by an education technology company in India. The raw dataset contains **21,256 candidate records** across **52 attributes**, capturing a wide range of information about each interviewee. For further details, refer to the full description of the variables provided in the Appendix.

The attributes in the original dataset span several categories:

- **Demographic information**: candidate name, age, gender, marital status, type of graduation, and geographic details.
- **Interview logistics**: mode of interview (mobile/in-person), pre-interview check status, and role location.
- **English fluency and confidence**: multiple assessments of the candidate's English fluency, confidence during self-introduction, topic discussion, PowerPoint presentation, and a sales role-play scenario. Confidence levels are recorded as categorical labels (e.g., "Impactful," "Guarded Confidence," "Nervous," "Low confidence").
- **Structured thinking**: evaluations of logical and structured thinking ability across different interview segments, including regional-language discussion, PPT-based questions, and a sales call pitch.
- **Regional fluency**: assessments of the candidate's communication skills in their regional language across multiple interview segments.
- **Work experience**: employment status, last fixed CTC (cost to company), experience duration in months, nature of prior work, and the type of role held.
- **Numerical scores**: aggregated scores for confidence (0–12), structured thinking (0–9), regional fluency (0–9), and a total composite score.
- **Interview verdict**: the final hiring decision, categorized as "Premium Select," "Borderline Select," "Select," "Reject," or "Borderline Reject."

The distribution of interview verdicts in the original dataset is: Select (9,607; 45.20%), Reject (4,250; 19.99%), Borderline Select (3,500; 16.47%), Premium Select (1,197; 5.63%), and Borderline Reject (150; 0.71%). The dataset is predominantly male (16,909 male vs. 4,347 female candidates).

B. Data Preprocessing

The original 52-column dataset contains many categorical and text-based attributes that are not directly compatible with the numerical decision matrices required by AHP and TOPSIS methods. A preprocessing pipeline was developed (implemented in a Jupyter notebook) to transform the raw data into a simplified, numerically scored format suitable for multi-criteria evaluation. The preprocessing steps are as follows:

Step 1: Criterion extraction and normalization. Six evaluation criteria were derived from the original attributes, each normalized to a 0–10 scale:

- **Confidence**: derived from the aggregated Confidence Score column (originally 0–12), linearly rescaled to 0–10 via score $=$ raw $\times\ 10/12$.
- **Communication**: derived from the Regional Fluency Score (originally 0–9), rescaled as score $=$ raw $\times\ 10/9$.
- **Technical Skills**: derived from the Structured Thinking Score (originally 0–9), rescaled as score $=$ raw $\times\ 10/9$.
- **Problem Solving**: also derived from the Structured Thinking Score, using the same rescaling.
- **Sales Ability**: mapped from the categorical "Confidence based on the sales scenario" field using a predefined mapping: "Impactful" $\rightarrow$ 9.0, "Guarded Confidence" $\rightarrow$ 7.0, "Nervous" $\rightarrow$ 5.0, and "Low confidence" $\rightarrow$ 3.0.
- **Experience**: converted from the categorical experience-in-months field into a numerical score using interval-based random sampling (e.g., Fresher $\rightarrow$ 5.5–6.5, 6–12 months $\rightarrow$ 6.5–7.5, 48+ months $\rightarrow$ 8.5–9.5), reflecting increasing value with longer experience.

Step 2: Outcome labeling. The five-level interview verdict was simplified into a binary classification: "Premium Select," "Borderline Select," and "Select" were mapped to *Selected*, while "Reject" and "Borderline Reject" were mapped to *Not Selected*. This binary label serves as a reference outcome for validating the MCDM rankings.

Step 3: Data cleaning. Rows with missing values were removed, and all numerical scores were rounded to one decimal place.

C. Processed Dataset

The resulting processed dataset contains **21,256 candidate records** with **8 attributes**: candidate name, six numerical evaluation criteria (confidence, communication, technical skills, problem solving, sales ability, and experience), and the binary final result. Table I summarizes the descriptive statistics of the six criteria.

TABLE I: Descriptive Statistics of Processed Evaluation Criteria

Criterion	Mean	Std	Min	Max
Confidence	6.77	2.53	0.8	10.0
Communication	5.94	3.27	0.0	10.0
Technical Skills	6.89	3.02	0.0	10.0
Problem Solving	6.89	3.02	0.0	10.0
Sales Ability	7.31	1.33	5.0	9.0
Experience	6.40	0.87	5.5	9.5

Of the 21,256 processed candidates, 14,304 (67.29%) are labeled as *Selected* and 6,952 (32.71%) as *Not Selected*. This distribution reflects the original recruitment process and provides a realistic context for the teaching case. The uniform 0–10 scale across all criteria ensures that the data can be directly used as input for the TOPSIS and Fuzzy TOPSIS decision matrices, while the pairwise comparison matrices for AHP and Fuzzy AHP are constructed separately by the decision maker based on subjective judgment of criterion importance.

V. Case Procedure

To support the practical application of multi-criteria decision-making techniques in recruitment evaluation, a web-based decision support system was developed and deployed at the following address: *https://ahp-calculator-puce.vercel.app/*

The platform guides users through a structured decision-making process using an interactive interface. It allows users to define a decision problem, specify evaluation criteria, input candidate alternatives, and generate ranking results using several Multi-Criteria Decision Making (MCDM) methods. The main objective of the system is to assist interviewers in performing a more objective and systematic candidate evaluation.

Figure 1 illustrates the overall workflow of the candidate selection process using the Hybrid Fuzzy AHP–TOPSIS system.

In this case study, the platform is applied to support **objective interview evaluation**. Instead of relying solely on subjective impressions, interviewers can formally define evaluation criteria and assess candidate performance using analytical decision models. This structured approach helps reduce bias and improves consistency in candidate assessment.

The system integrates several decision analysis methods:

- Analytic Hierarchy Process (AHP)
- Fuzzy Analytic Hierarchy Process (Fuzzy AHP)
- Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)
- Fuzzy TOPSIS
- Hybrid Fuzzy AHP–TOPSIS

Among these methods, this study focuses on the hybrid **Fuzzy AHP–TOPSIS** approach. Fuzzy AHP is used to determine the relative importance of evaluation criteria, while TOPSIS ranks candidates based on their closeness to the ideal solution. The use of fuzzy logic allows the model to represent uncertainty and linguistic judgments commonly used during interview assessments. I prefer this response

A. Phase 1: Defining the Decision Problem

The first phase requires users to define the decision objective, the evaluation criteria, and the candidate alternatives. This stage forms the foundation of the entire decision analysis process.

Users begin by specifying the **decision goal**, which describes the purpose of the evaluation. In the context of this case study, the goal is to **identify the most suitable candidate during the interview selection process**. Clearly defining the objective ensures that the decision model aligns with the recruitment objectives of the organization.

After defining the decision goal, users must determine the **evaluation criteria**. These criteria represent the factors used to assess candidate performance during interviews. Typical interview evaluation criteria may include technical competency, communication skills, work experience, problem-solving ability, teamwork capability, or cultural fit. The system requires at least two criteria in order to perform the analysis.

The platform provides two approaches for defining evaluation criteria.

1) Manual Criteria Entry: Users may manually define the evaluation criteria directly within the system interface. Each criterion is assigned a descriptive name that reflects the aspect of candidate performance being evaluated.

In addition, each criterion must be categorized according to its evaluation type:

- **Benefit criteria**: Higher values indicate better candidate performance (e.g., technical skills, leadership ability, or communication effectiveness).
- **Cost criteria**: Lower values indicate better performance (e.g., expected salary, training cost, or project risk).

Users may add multiple criteria dynamically using the *Add Criterion* feature provided in the interface. This flexibility allows the system to accommodate different recruitment scenarios and evaluation frameworks.

2) Dataset Import: Alternatively, users may upload a dataset containing predefined criteria and candidate information. When a dataset is provided, the system automatically extracts the relevant fields and populates the decision structure accordingly.

This functionality is particularly useful when organizations maintain structured recruitment data or when evaluating a large number of candidates. By importing existing datasets, the platform reduces manual data entry and helps ensure consistency in the evaluation process.

Once the decision goal, criteria, and candidate alternatives have been defined, the system proceeds to the next phase. In this phase, users perform pairwise comparisons between evaluation criteria using the Fuzzy AHP method in order to determine their relative importance within the interview decision-making process.

B. Phase 2: Pairwise Comparison of Criteria Using Fuzzy AHP

After defining the decision goal, criteria, and candidate alternatives, the next phase involves determining the relative importance of each evaluation criterion. This is achieved

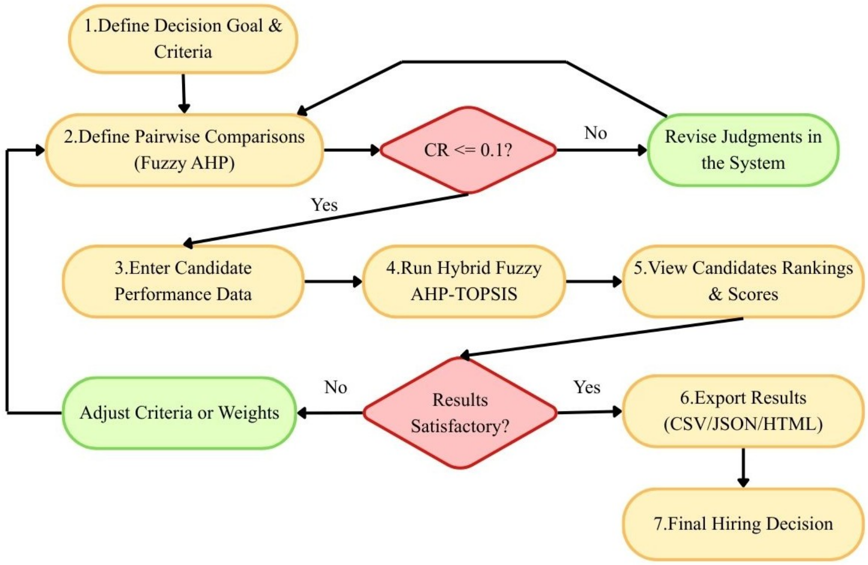

Fig. 1: Workflow of the Hybrid Fuzzy AHP–TOPSIS Candidate Selection Process.

through the **Fuzzy Analytic Hierarchy Process (Fuzzy AHP)** method, which requires users to perform pairwise comparisons between criteria.

In this phase, the system asks users to compare two criteria at a time and evaluate how important one criterion is relative to the other. Instead of assigning direct numerical weights, the user expresses judgments using a predefined importance scale. This approach allows the system to capture human judgment in a structured way.

The platform adopts the commonly used AHP importance scale:

- 1 – Equal importance
- 3 – Moderate importance
- 5 – Strong importance
- 7 – Very strong importance
- 9 – Extreme importance
- 1/3 – 1/9 – Indicates that the compared criterion is less important

For each pair of criteria, the user selects a value from the scale to represent their judgment. For example, if *Technical Skills* are considered strongly more important than *Communication Skills*, the user may assign a value of 5 to represent this relationship.

In addition to the comparison value, the system also allows users to specify their **confidence level** in the judgment. This feature helps represent the uncertainty that often occurs during human evaluation. The selected confidence level determines the range of triangular fuzzy numbers used to model the comparison.

Through this process, the system constructs a **pairwise comparison matrix** for all criteria. The Fuzzy AHP algorithm then processes this matrix to compute the fuzzy weights representing the relative importance of each criterion.

The resulting weights will be used in the next phase, where candidate alternatives are evaluated and ranked using the TOPSIS method.

C. Phase 3: Entering Candidate Performance Data

After the criteria weights have been determined using the Fuzzy AHP method, the next phase involves evaluating each candidate according to the defined criteria. This step forms the basis for the ranking process performed using the **TOPSIS** method.

In this phase, users provide performance scores for each candidate across all evaluation criteria. The system applies the criteria weights obtained from the previous phase automatically during the ranking process.

The platform uses a numerical scoring scale from **1 to 10**, where higher values represent better performance on a given criterion. For example, a candidate who demonstrates excellent technical knowledge may receive a higher score in the *Technical Skills* criterion, while a candidate with weaker communication abilities may receive a lower score in the *Communication* criterion.

The system provides two methods for entering candidate performance data.

1) Manual Data Entry: Users may manually input the evaluation scores directly into the system. For each candidate, a score must be assigned for every evaluation criterion. This approach allows interviewers to record their assessments immediately after conducting interviews.

Manual entry is particularly useful when the number of candidates is relatively small or when interviewers prefer to directly input evaluation results during the assessment process.

2) Dataset Import: Alternatively, users may load a dataset that contains pre-recorded candidate performance data. When a compatible dataset is uploaded, the system automatically populates the evaluation table with the corresponding scores for each candidate and criterion.

Using a dataset significantly reduces manual input and is especially beneficial when evaluating a large number of candidates or when assessment data has already been collected in advance.

Once the candidate performance data has been completed, the system proceeds to the final phase, where the **TOPSIS** method is applied to compute the ranking of candidates based on their relative closeness to the ideal solution.

D. Phase 4: Results and Analysis

The final phase of the system presents the decision results generated using the Hybrid Fuzzy AHP–TOPSIS method. After the criteria weights are determined and candidate performance data are evaluated, the system computes the final ranking of candidates based on the closeness coefficient obtained from the TOPSIS calculation. This phase allows users to interpret the results, compare candidates, and export the analysis for reporting purposes.

1) Candidate Ranking: The system generates a ranked list of candidates according to their closeness coefficient values. The closeness coefficient represents how close a candidate is to the ideal solution and how far they are from the negative ideal solution. Candidates with higher closeness coefficient values are considered more suitable for the position.

The ranking interface displays each candidate along with their corresponding score and ranking position. This provides decision-makers with a clear overview of the best-performing candidates based on the weighted evaluation criteria.

2) Candidate Comparison View: To provide deeper insight into the evaluation results, the system includes a comparison view that visually displays the performance of top-ranked candidates across all criteria. In this view, each candidate's scores for confidence, communication, technical skills, problem solving, sales ability, and experience are presented side by side.

This comparison allows decision-makers to easily identify strengths and weaknesses among candidates and supports more transparent hiring discussions.

3) Detailed Calculation Explanation: To improve transparency and understanding of the decision-making process, the system provides a detailed breakdown of the calculation steps used to produce the final results. This includes:

- The fuzzy criterion weights obtained from the Fuzzy AHP process
- The pairwise comparison matrix used to determine the relative importance of criteria
- The alternative performance data entered for each candidate
- The distance measures to the ideal best and ideal worst solutions
- The computed closeness coefficient values for each candidate

By presenting these intermediate calculations, users can verify how the final ranking was generated and ensure the decision-making process remains transparent and objective.

4) Result Export and System Reset: The system also provides several options for exporting the evaluation results. Users can export the results in multiple formats, including CSV, JSON, and HTML files. This allows the data to be used for documentation, reporting, or further analysis.

Additionally, a system reset option is available, allowing users to start a new evaluation process. This feature enables the system to be reused for different hiring scenarios or datasets without needing to manually clear previous data.

Overall, this phase ensures that the final decision results are clearly presented, easily interpretable, and available for documentation and further review.

VI. Conclusion

This paper presented a teaching case on the application of a Hybrid Fuzzy AHP–TOPSIS approach to support multi-criteria candidate selection in recruitment processes. The study began by introducing the challenges associated with subjective and unstructured hiring decisions [2], followed by a review of established MCDM methods including AHP [3], Fuzzy AHP [5], [10], TOPSIS [6], and Fuzzy TOPSIS [7]. The mathematical foundations of each method were formally described, and a hybrid framework that combines Fuzzy AHP for criteria weighting with Fuzzy TOPSIS for alternative ranking was proposed [12], [13].

To demonstrate the practical application of the framework, a web-based decision support system was developed and deployed. The platform guides users through a structured four-phase workflow: defining the decision problem, performing pairwise comparisons using Fuzzy AHP, entering candidate performance data, and generating ranked results with detailed calculation transparency. The system supports both manual data entry and dataset import, making it adaptable to various recruitment scenarios.

The case study was applied to a real-world interview selection dataset sourced from Kaggle, consisting of 21,256 records across 52 variables. Through preprocessing, six representative evaluation criteria were derived, and candidate outcomes were classified into a binary selection label. This dataset served as the basis for demonstrating the end-to-end evaluation process using the hybrid method.

By integrating fuzzy logic into the decision-making process, the proposed framework effectively captures the uncertainty and subjectivity inherent in human judgment during candidate evaluation [8]. The use of triangular fuzzy numbers allows evaluators to express their assessments with varying degrees of confidence, while the consistency ratio ensures the reliability of pairwise comparisons [3].

This teaching case equips learners with practical skills in structured decision analysis, including defining decision goals, weighting evaluation criteria, ranking alternatives, and interpreting quantitative results. The accompanying web application and source code provide an accessible and interactive learning environment for exploring MCDM techniques in a real-world context.

Future work may extend the framework by supporting group decision-making with multiple evaluators, integrating machine learning to automate criteria weighting from historical hiring data [14], or incorporating AI-based components into the pipeline. Recent work has shown that AI complements Fuzzy AHP–TOPSIS in complex settings [15], while LLMs automate profile analysis and score extraction from unstructured text [16], significantly boosting system scalability.

REFERENCES

[1] Dursun, M., & Karsak, E. E. (2010). A fuzzy MCDM approach for personnel selection. *Expert Systems with Applications*, 37(6), 4324–4330. https://doi.org/10.1016/j.eswa.2009.11.067

[2] Radbruch, J., & Schiprowski, A. (2024). Interview sequences and the formation of subjective assessments. *The Review of Economic Studies*, 92(2), 1226–1256. https://doi.org/10.1093/restud/rdae039

[3] Saaty, T. L., & Vargas, L. G. (2012). Models, Methods, Concepts & applications of the Analytic Hierarchy Process. In International Series in Operations Research & Management Science (Second, Vol. 175). Springer. https://doi.org/10.1007/978-1-4614-3597-6

[4] Ishizaka, A., & Labib, A. (2011). Review of the main developments in the analytic hierarchy process. *Expert Systems with Applications*, 38(11), 14336–14345. https://doi.org/10.1016/j.eswa.2011.04.143

[5] Chang, D. (1996). Applications of the extent analysis method on fuzzy AHP. European Journal of Operational Research, 95(3), 649–655. https://doi.org/10.1016/0377-2217(95)00300-2

[6] Tzeng, G.-H., & Huang, J.-J. (2011). Multiple attribute decision making methods and applications. Taylor & Francis Group. http://www.taylorandfrancis.com

[7] Chen, C. (2000). Extensions of the TOPSIS for group decision-making under fuzzy environment. Fuzzy Sets and Systems, 114(1), 1–9. https://doi.org/10.1016/s0165-0114(97)00377-1

[8] Kabir, G., & Hasin, M. A. A. (2011). COMPARATIVE ANALYSIS OF AHP AND FUZZY AHP MODELS FOR MULTICRITERIA INVENTORY CLASSIFICATION. In Bangladesh University of Science and Technology (BUET), International Journal of Fuzzy Logic Systems (IJFLS): Vol. Vol.1 (Issue No.1).

[9] Xiao, Z., Chen, L., & Zhong, B. (2010). A model based on rough set theory combined with algebraic structure and its application: Bridges maintenance management evaluation. Expert Systems With Applications, 37(7), 5295–5299. https://doi.org/10.1016/j.eswa.2010.01.010

[10] Buckley, J. (1985). Fuzzy hierarchical analysis. Fuzzy Sets and Systems, 17(3), 233–247. https://doi.org/10.1016/0165-0114(85)90090-9

[11] Zadeh, L. A. (1965). Fuzzy sets. *Information and Control*, 8(3), 338–353. https://doi.org/10.1016/S0019-9958(65)90241-X

[12] Kutlu, A. C., & Ekmekçioğlu, M. (2011). Fuzzy failure modes and effects analysis by using fuzzy TOPSIS-based fuzzy AHP. Expert Systems With Applications, 39(1), 61–67. https://doi.org/10.1016/j.eswa.2011.06.044

[13] Sun, C. (2010). A performance evaluation model by integrating fuzzy AHP and fuzzy TOPSIS methods. Expert Systems With Applications, 37(12), 7745–7754. https://doi.org/10.1016/j.eswa.2010.04.066

[14] Faliagka, E., Tsakalidis, A., & Tzimas, G. (2012). An integrated e-recruitment system for automated personality mining and applicant ranking. *Internet Research*, 22(5), 551–568. https://doi.org/10.1108/10662241211271545

[15] Kılıç, C., Balcı, O., & Gül, G. (2025). Unraveling Decision-Making Complexity: Artificial Intelligence versus Fuzzy AHP-TOPSIS [Research Article]. Journal of Emerging Economies and Policy, 10(1), 227–247. http://dergipark.org.tr/joeep (Original work published 2024)

[16] Hoque, S., Karim, A. A. J., Alam, Md. G. R., & Nirjhar Gope. (2026). When LLM meets Fuzzy-TOPSIS for Personnel Selection through Automated Profile Analysis [Journal-article]. BRAC University Department of Computer Science and Engineering, 4. https://arxiv.org/abs/2601.22433v1

Appendix A
Kaggle Dataset Description

TABLE II: Variable Description of the Kaggle Interview Selection Dataset

Variable	Description
Name	A character variable indicating the anonymized name of the candidate.
Age	An integer variable indicating the age of the candidate.
Gender	A character variable indicating the gender of the candidate (Male/Female).
Type of Graduation/Post Graduation	A character variable indicating the candidate's highest educational qualification (e.g., B.E/B-Tech, MBA, BSc/MSc).
Marital status	A character variable indicating whether the candidate is married or unmarried.
Mode of interview given by candidate?	A character variable indicating the interview mode (e.g., Mobile, In-person).
Pre Interview Check	A character variable indicating the outcome of the pre-interview screening.
Fluency in English based on introduction	A character variable describing the candidate's English fluency during self-introduction.
Confidence based on Introduction (English)	A character variable rating the candidate's confidence during the English introduction segment.
Confidence based on the topic given	A character variable rating confidence during a topic-based discussion.
Confidence Based on the PPT Question	A character variable rating confidence during a PowerPoint presentation question.
Confidence based on the sales scenario	A character variable rating confidence during a sales role-play scenario.
Structured Thinking (In regional only)	A character variable evaluating logical and structured thinking in the candidate's regional language.
Structured Thinking Based on the PPT Question	A character variable evaluating structured thinking during the PPT question segment.
Structured Thinking (Call pitch)	A character variable evaluating structured thinking during the sales call pitch.
Regional fluency based on the topic given	A character variable assessing regional language fluency during the topic discussion.
Regional fluency Based on the PPT Question	A character variable assessing regional fluency during the PPT question segment.
Regional fluency based on the sales scenario	A character variable assessing regional fluency during the sales scenario.
Does the candidate has mother tongue influence?	A character variable (Yes/No) indicating whether the candidate's mother tongue influences their English.
Has acquaintance in Company?	A character variable (Yes/No) indicating whether the candidate knows someone in the company.
Candidate Status	A character variable describing the candidate's career level (e.g., Fresher, Experienced, Lateral).
Last Fixed CTC (lakhs)	A character variable indicating the candidate's last fixed annual compensation in lakhs.
Currently Employed	A character variable (Yes/No) indicating current employment status.
Experience (in months)	A character variable indicating the candidate's work experience duration category.
Nature of work	A character variable describing the type of prior work experience (e.g., Sales, Tech, Fresher).
Type of Role	A character variable indicating the type of role previously held (e.g., Individual Contributor, Fresher).
Slides submitted in PPT	A character variable indicating how many PPT slides the candidate prepared.
Call-pitch Elements used	A character variable listing the elements used during the sales call pitch scenario.
Sales objection handling (3 columns)	Three character variables recording the candidate's responses to specific sales objection scenarios.
Role acceptance	A character variable indicating the candidate's willingness to accept the offered role.
Interview Verdict	A character variable indicating the final hiring decision (Premium Select, Borderline Select, Select, Reject, Borderline Reject).
Candidate is willing to relocate	A character variable indicating the candidate's relocation willingness.
Role Location	A character variable indicating the assigned role location.
Comments	A character variable containing free-text interviewer comments.
RedFlags Comments in Interview	A character variable listing any red flags noted during the interview.
Confidence Score (5 columns)	A numeric variable (0–12) representing the aggregated confidence score across all segments.
Structured Thinking Score (4 columns)	A numeric variable (0–9) representing the aggregated structured thinking score.
Regional Fluency Score (4 columns)	A numeric variable (0–9) representing the aggregated regional fluency score.
Total Score	A numeric variable representing the total composite score of the candidate.
Whether joined the company or not	A character variable indicating whether the selected candidate ultimately joined the company.